

Operators

Learning Objectives

- Use mathematical operators for calculations
- Apply comparison and logical operators
- Combine operators in expressions and conditions

Engage (5-7 minutes)

Think of a shopkeeper calculating bills—adding prices, comparing amounts, checking if customers have enough money. Operators in Scratch do math and comparisons just like that. Today we learn to make our programs calculate and make smart comparisons.

Explore (10-12 minutes)

Use the Operators category. Try “add,” “subtract,” “multiply,” and “divide” with numbers. Create a variable “answer” and use “set answer to $5 + 3$.” Try comparison operators like “greater than” and “less than.” Combine two conditions with “and” and “or” blocks.

Explain (10-12 minutes)

Operators are tools for calculations and comparisons. Arithmetic operators do math (add, subtract, multiply, divide). Comparison operators check relationships (greater than, less than, equals). Logical operators combine conditions (and, or, not). These work inside conditions and variable assignments.

Elaborate (8-10 minutes)

Build a calculator project where clicking number buttons adds values to a total. Create a quiz where the correct answer is calculated using operators. Use comparisons to check if a score is high enough to win. Like a weighing scale that compares two quantities, operators help your programs compare and calculate.

Evaluate (5-8 minutes)

Exit ticket: What does “ $15 / 3$ ” equal in Scratch? Write a condition checking if a variable “age” is greater than 10. How would you use “and” to combine two conditions?